

Worksheet 8

September 30, 2026

Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

1. Domain and a hole

Let

$$R(x) = \frac{x - 3}{x^2 - 9}.$$

(a) Factor the denominator. Give the original domain in interval notation.

(b) Simplify R while keeping its restrictions. Find the hole as an ordered pair.

(c) Find all x - and y -intercepts, or state that none exist.

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2. Asymptotes and limits

Let

$$R(x) = \frac{2x + 1}{x - 1}.$$

(a) Find the vertical and horizontal asymptotes. Explain each.

(b) Find $\lim_{x \rightarrow 1^-} R(x)$ and $\lim_{x \rightarrow 1^+} R(x)$.

(c) Find $\lim_{x \rightarrow -\infty} R(x)$ and $\lim_{x \rightarrow \infty} R(x)$.